Clinical Study Summary Report (CSSR)
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1. Administrative & Study Identification

Full Study Title: Phase 1 Study to Evaluate the Mass Balance, Pharmacokinetics, Metabolism, Excretion and Absolute Bioavailability of Tuvusertib (M1774) Containing Microtracer 14C Tuvusertib in Participants With Advanced Solid Tumors **Short Title/Acronym:** DDRIVER Solid Tumors 303
Protocol Number: MS201924_0003 (EU Trial Number: 2022-502940-10-00)
NCT Number: NCT06308263
Sponsor Name: Merck Healthcare KGaA, Darmstadt, Germany
Investigational Product: Tuvusertib (M1774) and [14C]Tuvusertib microtracer
Phase of Development: Phase 1
Medical Monitor: Clinical Study Director, Merck Healthcare KGaA

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the mass balance to determine drug-related entities present in circulation and excreta, provide a comprehensive understanding of biotransformation pathways and clearance mechanisms, and assess the extent of absolute bioavailability (ABA) of tuvusertib. **Secondary Objectives:** To evaluate the safety profile, overall tolerability, and ongoing preliminary anti-tumor activity in participants entering the extension phase.

2.2 Background & Rationale Tuvusertib (M1774) is a potent, orally available inhibitor of Ataxia telangiectasia mutated and Rad3-related (ATR) kinase, exhibiting anti-proliferative effects in cancer models. To support its continued clinical development, it is a regulatory necessity to fully understand its Absorption, Distribution, Metabolism, and Excretion (ADME) profile. This mass balance and absolute bioavailability study will utilize a 14C-radiolabeled microtracer to precisely map biotransformation pathways and clearance mechanisms in human subjects with advanced solid tumors.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Single sequence, 2-period open-label study) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target N: 12 evaluable patients **Number of Sites:** Global multicenter trial **Estimated**

Study Start: 25-Mar-2024 **Estimated Primary Completion:** 23-Mar-2026

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically proven advanced solid tumors that are considered appropriate for treatment in Period 2 of this study, for which no effective standard therapy exists, or standard therapy has failed or cannot be tolerated. 2. Eastern Cooperative Oncology Group Performance Status (ECOG PS) ≤ 1. 3. Have evaluable disease according to Response Evaluation Criteria in Solid Tumors (RECIST) 1.1 at Screening.

4.2 Exclusion Criteria 1. Uncontrolled or poorly controlled arterial hypertension, symptomatic congestive heart failure (NYHA $\ge$ Class III), uncontrolled cardiac arrhythmia, calculated QTcF > 480 msec, unstable angina pectoris, myocardial infarction, or significant vascular disease within 180 days. 2. Presence of toxicities due to prior anticancer therapies that do not recover to $\le$ Grade 1 (with the exception of non-safety risks like ongoing Grade 2 alopecia). 3. Treatment with live or live attenuated vaccine within 30 days of dosing, or participation in a study involving administration of 14C-labeled compound(s) within the last 6 months.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Standard oral Tuvusertib dosing combined with a [14C]Tuvusertib microdose bolus injection during Period 1/1a. **Route of Administration:** Oral (unlabeled drug) and Intravenous (microtracer). **Duration of Treatment:** Single sequence Period 1/1a for ADME profiling, followed by an optional extension phase (Period 2) where participants will receive continuous oral tuvusertib until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard palliative and supportive care medications for symptom management. **Prohibited:** Strong CYP inhibitors/inducers (based on ADME restrictions), live or live-attenuated vaccines within 30 days, and any other 14C-labeled radiopharmaceuticals.

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6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Day -28 to -1 | Informed Consent, RECIST 1.1 imaging, ECG, ECOG Assessment | | **Period 1/1a (Baseline)** | Day 1 to Day 7+ | Oral and IV Dose Administration, Intensive blood, urine, and fecal sampling for mass balance/PK | | **Period 2 (Extension)** | Cycle 1, Day 1 onwards | Continuous oral

dosing, Safety Labs, Efficacy Assessment (RECIST) | | **End of Study** | +30 Days post-last dose | Final Vital Signs, Exit Interview, IP Accountability, Safety follow-up |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Extent of Absolute Bioavailability (ABA) and determination of the mass balance, pharmacokinetics (PK), metabolism, and elimination of 14C-tuvusertib. **Measurement Method:** Liquid scintillation counting and Accelerator Mass Spectrometry (AMS) to measure radioactivity in whole blood, plasma, urine, and feces. Evaluation of standard PK parameters (C_{max}, AUC).

7.2 Secondary & Exploratory Endpoints Incidence and severity of treatment-emergent adverse events (TEAEs). Objective Response Rate (ORR) and Progression-Free Survival (PFS) per RECIST 1.1 for patients entering the Period 2 extension phase.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via laboratory parameters, vital signs, physical examinations, and 12-lead ECGs throughout the intensive ADME phase and extension. **Serious Adverse Events (SAEs):** Must be reported to the sponsor within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal safety review committee governing dose escalations and ADME clearance.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: * *PK/Mass Balance Set*: All enrolled participants who receive the microdose and have sufficient samples collected. * *Safety Set*: All participants who receive at least one dose of study medication. **Sample Size Justification:** A sample size of $N=12$ is practically and statistically adequate for a human ADME/mass balance Phase 1 study aiming to characterize general PK profiles rather than test a comparative hypothesis. **Handling of Missing Data:** PK concentrations below the limit of quantification (BLQ) will be treated as zero for descriptive statistics. Standard non-compartmental analysis (NCA) models will be used.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Rigorous onsite monitoring during the Period

1/1a sampling window to ensure accurate timing and handling of radioactive biological samples. **Audit Trail:** Validated eCRF capture, radiolabeled material accountability logs, and strictly controlled centralized laboratory manifests.

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11. Study Identifiers & Governance

Primary ID: MS201924_0003 **Registry Number:** NCT06308263 **Sponsor/ Lead Organization:** Merck Healthcare KGaA, Darmstadt, Germany **Study Title:** DDRIVER Solid Tumors 303 **Official Regulatory Title:** Phase 1 Study to Evaluate the Mass Balance, Pharmacokinetics, Metabolism, Excretion and Absolute Bioavailability of Tuvusertib (M1774) Containing Microtracer 14C Tuvusertib in Participants With Advanced Solid Tumors

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12. Clinical Framework (Solid Tumor Specific)

Primary Condition: Advanced Solid Tumors **Diagnosis Threshold:** Histologically proven malignancy with evaluable disease per RECIST 1.1 **Medical Methodology:** ATR Kinase Inhibition Therapy (Tuvusertib) **Optimization Target:** Pharmacokinetic and Biotransformation profiling (ADME)

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13. Study Procedures & Methodology

Management Pathway: Intensive ADME and PK profiling pathway for early-phase oncology agents **Education Protocol (HCP):** Site initiation visits covering radioactive microtracer handling, strict biological sampling schedules, and AMS processing **Education Protocol (Patient):** Informed consent detailing the intensive urine/feces collection requirements and exposure to trace radiolabeled substances **Quality Audit Mechanism:** Real-time query generation for PK sampling deviations and strict radiolabel accountability monitoring

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14. Observation & Follow-Up Schedule

Total Duration: Dependent on individual participant response (Period 1 ADME is short-term; Period 2 continues until progression) **Onsite Visit Cadence:** Daily/Intensive during Period 1/1a for strict PK sampling; Transitioning to standard Oncology Q3W/Q4W cycles for Period 2 **Remote Monitoring Frequency:** N/A (Intensive phase requires entirely onsite biological collection) **Checklist Review Interval:** Regular tumor assessments (RECIST 1.1) every 8-12 weeks during the extension phase

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]